

Overfit-Aware Signals

Evaluating cross-sectional equity factors with DSR and PBO

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What it is

Most backtests report one Sharpe ratio and stop. Under multiple testing that number is easy to inflate. This package runs three real cross-sectional signals, 12-1 momentum, one-month reversal, and low-vol, through purged CV, combinatorial purged CV, Deflated Sharpe Ratio (Bailey and Lopez de Prado, 2014), and Probability of Backtest Overfitting via CSCV. The product is a per-signal PASS or FAIL verdict under fixed thresholds, not a vanity Sharpe chart.

Results (live universe, Jul 2026 run)

50 liquid US equities, monthly rebalance, 2005-01 to 2026-07, long-only top-5, 10 bps costs. DSR threshold ≥ 0.95 , PBO threshold ≤ 0.05 . PBO is set-level, same across rows. `n_trials = 3`.

signal	Sharpe	CAGR	median OOS Sharpe	DSR	PBO	verdict
momentum	0.937	0.187	1.039	0.999	0.031	PASS
reversal	0.808	0.186	0.882	0.998	0.031	PASS
lowvol	0.816	0.103	0.836	0.995	0.031	PASS

Methods

- Purged K-Fold with embargo (AFML ch. 7): purge on look-back-window overlap, not only on labels
- CPCV: $S = 8$, $k = 2$, giving 28 out-of-sample Sharpe paths
- $DSR = PSR(SR_0)$ using non-excess kurtosis (pandas `kurt + 3`)
- PBO via CSCV, reject if $PBO > 0.05$

Caveat

Universe uses current constituents, not a point-in-time survivorship-free panel. Absolute Sharpes are likely optimistic. The DSR and PBO machinery is still valid for within-panel multiple-testing correction. Not trading advice.

Reproduce

```
pip install -e ".[dev]"
python -m overfit_aware_signals run
```

Repo

Endokelp_Projects / PythonProjects / overfit-aware-signals

Primary sources: Bailey and Lopez de Prado (2014), Deflated Sharpe Ratio; Bailey et al., Probability of Backtest Overfitting and CSCV; Lopez de Prado, Advances in Financial Machine Learning, ch. 7. Positioned as claim-filtering tooling, the same class of checks used in 2026 LLM strategy-to-code benchmarks, not an AFML tutorial clone.